Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus
Golden Country Homes, Alangilan Batangas City, Batangas, Philippines 4200
Tel Nos.: (+63 43) 425-0139 local 2222 / 2223
E-mail Address: cics.alangilan@g.batstate-u.edu.ph | Website Address: http://www.batstate-u.edu.ph

College of Informatics and Computing Sciences

PROJECT PROPOSAL
OOP 101 – OBJECT-ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026–2027

I. TEAM INFORMATION

Team Name: Saksi ni Java			
Team Number: 2		Section: IT - 2108	
No.	**SR-Code**	**Name**	**GitHub Username**
1	25-04083	Agda, Merald M.	MeraldAgda
2	25-01488	De Chavez, Kate Darrize C.	dechavezkatkat06-code
3	25-04986	Marquinez, Erick Angelo De Los Reyes	ERICK ANGELO MARQUINEZ
4	25-04926	Palma, Gil Lorence B.	Tsuguosenbei

II. PROJECT INFORMATION

Project Title
STAYHUB AIRBNB BOOKING SYSTEM

Project Background

Finding and managing short-term accommodations can be challenging when property information, availability, and reservation details are handled manually or are not organized in one system. This project addresses the need for a simple and organized way to browse properties, check their availability, calculate rental costs, manage reservations, and cancel an existing reservation when necessary.

The problem is mainly experienced by guests looking for accommodations and property owners managing property listings. Guests need an easy way to view available properties, make reservations, and cancel reservations, while property owners need a convenient way to add new property listings.

Solving this problem is important because a booking system can organize property and reservation information, reduce errors in the booking process, and make the process more convenient for both guests and property owners. Through StayHub, the project also demonstrates how object-oriented programming can be applied to create a practical and organized Java application.

Project Objectives

General Objective

To develop a Java-based console application that simulates an Airbnb-style property booking system using appropriate object-oriented programming concepts.

Specific Objectives
The project aims to:

- To allow guests to browse and view details of available and listed properties.
- To enable guests to select a property and create a reservation based on availability and booking requirements.
- To allow guests to cancel an existing reservation and restore the property's availability.
- To calculate the total rental cost according to the selected property type and length of stay.

- To provide a property management portal where authorized property owners can add new property listings.
- To store and display active reservations with their corresponding booking details.
- To demonstrate the proper application of abstraction, encapsulation, inheritance, polymorphism, exception handling, and data handling.

Target Users

Target User	How They Will Use the System
Guests	Browse available properties, view property details, select a property, enter booking information, and receive a reservation receipt.
Property Owners/Hosts	Access the property management portal using the required PIN and add new property listings such as apartments, villas, and shared rooms.

Proposed Features

Feature	Description
Browse Properties	Displays all listed properties and their details, including location, nightly rate, special features, and booking status.
Property Booking	Allows guests to select an available property, enter booking information, and receive a reservation receipt.
Reservation Cancellation	Allows guests to cancel an active reservation using its reservation ID. The property's availability is restored and the reservation is removed from the active list.
Rental Calculation	Computes the total rent based on property type and number of nights, including applicable fees and the villa long-stay discount.
Property Management	Uses a PIN-protected portal that allows property owners to add new apartment, villa, or shared-room listings.
View Active Reservations	Displays the details of all reservations currently stored in the system.

III. JAVA AND OOP CONCEPTS

OOP Concept	How It Will Be Used
Classes and Objects	Classes such as Property, ApartmentProperty, VillaProperty, SharedRoomProperty, and Reservation model the main parts of the booking system.
Methods	Methods organize operations such as browsing properties, booking a property, cancelling a reservation, adding a listing, and viewing reservations.
Encapsulation	Property and Reservation attributes are private and accessed through public getter and setter methods where appropriate.
Inheritance	ApartmentProperty, VillaProperty, and SharedRoomProperty inherit common attributes and behavior from the abstract Property class.
Polymorphism	Different property subclasses are stored as Property objects and use overridden methods such as calculateTotalRent() and displayPropertyDetails().

Abstraction	The Bookable interface defines common booking actions, including checking availability, confirming a reservation, and cancelling a reservation. The abstract Property class provides shared data and behavior.
Exception Handling	Custom exceptions handle unavailable properties and invalid bookings, while try-catch blocks handle invalid numeric input and other runtime errors.
Arrays/Data Handling	ArrayList collections store property listings and active reservations for browsing, searching, adding, cancelling, and displaying data.

IV. PROPOSED CLASSES

Class	Purpose	Main Attributes	Main Methods
Bookable (Interface)	Defines common actions for reservable properties.	None	calculateTotalRent(), checkAvailability(), confirmReservation(), cancelReservation()
Property (Abstract Class)	Common parent class for all property types.	propertyId, title, location, baseNightlyRate, isAvailable	Getters/setters, checkAvailability(), confirmReservation(), cancelReservation(), displayPropertyDetails()
ApartmentProperty	Represents an apartment listing.	floorLevel, hasElevator	calculateTotalRent(), displayPropertyDetails()
VillaProperty	Represents a villa listing with special stay requirements.	hasPrivatePool, minNightsRequired	getMinNightsRequired(), calculateTotalRent(), displayPropertyDetails()
SharedRoomProperty	Represents a shared-room listing.	maxRoommates	calculateTotalRent(), displayPropertyDetails()
Reservation	Stores completed booking details.	reservationId, property, guestName, nights, totalAmount	getReservationId(), getProperty(), getGuestName(), displayReceipt()
Main	Controls the application flow and system operations.	properties, reservations, reservationCounter, HOST_PIN	main(), browseProperties(), bookProperty(), cancelReservation(), addProperty(), viewReservations(), seedSampleData()

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V. PROGRAM FLOW

Start
↓
Initialize & Seed Sample Data (P101, P102, P103)
↓
Display Main Menu (Options 1–6)
↓
Validate Input (Numeric 1–6)
├── Invalid Input ──> Display Error Message ──> Return to Main Menu
└── Valid Input ──> Route to Selected Feature (Options 1–6)
├── Option 1: Browse Properties
├── Option 2: Book Property (Guest Portal)
├── Option 3: Cancel Reservation (Guest Portal)
├── Option 4: Add New Listing (Host Portal)
├── Option 5: View Active Reservations
└── Option 6: Exit System
↓
Is Choice == 6 (Exit)?
├── No ──> Return to Main Menu
└── Yes ──> Close Resources & Terminate Program
↓
Exit

VI. TEAM RESPONSIBILITIES

Member	Assigned Tasks
Gil Lorence Palma	Team Leader
Kate Darrize De Chavez	Documentation Leader
Erick Angelo Marquinez	System / Design Leader
Merald Agda	Lead Programmer

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and requirements	Completed project proposal and defined system requirements
Week 2	System and class design	Initial class design and program flow
Week 3	Initial coding	Basic classes, interface, and abstract class implementation
Week 4–5	Feature development	Working property browsing, booking, cancellation, reservation, and listing features
Week 6	OOP implementation and integration	Completed abstraction, encapsulation, inheritance, polymorphism, and exception handling
Week 7	Testing and debugging	Tested application with corrected errors and validated input handling
Week 8	Documentation and presentation	Final source code, documentation, and project presentation

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VIII. EXPECTED OUTPUT

The team expects to produce a functional Java console application that simulates a basic Airbnb-style property booking system. The completed application will allow guests to browse property listings, make reservations, cancel active reservations, and view property information. Authorized property owners will be able to add new listings.

The application will calculate rental costs based on property type and length of stay, maintain a list of active reservations, and restore a property's availability when a reservation is cancelled. The final output will demonstrate the appropriate use of Java and object-oriented programming concepts through a practical and organized system design.

IX. PROJECT LIMITATIONS

- The application is console-based and does not include a graphical user interface.
- Property and reservation data are stored only during program execution and are not connected to a database.
- The system does not process actual online payments or financial transactions.
- The system uses a simple access PIN and does not implement full user account registration or advanced authentication.
- The current version does not include date-based booking schedules, reservation history, or modification of existing reservations.
- The project focuses on core features needed to demonstrate OOP concepts and does not aim to replicate all features of a commercial Airbnb platform.

X. GITHUB REPOSITORY

The repository will contain the source code, README, documentation, UML/class diagram, screenshots, and other required project files.

Repository Name: STAYHUB-AIRBNB-BOOKING-SYSTEM

Link to Repository: https://github.com/Tsuguosenbei/STAYHUB-AIRBNB-BOOKING-SYSTEM

XI. APPROVAL

Prepared by:

Name	Signature
Agda, Merald M.	
De Chavez, Kate Darrize C.	
Marquinez, Erick Angelo De Los Reyes	
Palma, Gil Lorence B.	